

Phishing Email Analysis Report

Email File: sample-1034.eml converted to email_body_clean.txt

Analysis Date: 22-Oct-2025

Email file link: https://github.com/rf-peixoto/phishing_pot/blob/main/email/sample-1034.eml

CONVERSION OF EML TO TXT

1 Extract HTML content from the email

If your email (sample-1034.eml) contains HTML formatting, first extract that part:

```
grep -a -A 2000 "<html" sample-1034.eml > email_body.html
```

- `grep -a` → reads the file as text (even if binary parts exist)
 - `-A 2000` → captures 2000 lines *after* `<html>`
 - The output (`> email_body.html`) saves the HTML part separately
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2 Convert HTML to safe text (no links executed)

Use the lynx command-line browser to render it safely into plain text:

```
lynx -dump -nolist email_body.html > email_body.txt
```

- `lynx -dump` → prints the webpage text content
- `-nolist` → removes numbered hyperlinks
- Output: `email_body.txt` (plain text version)

If you didn't have lynx, you installed it using:

```
sudo apt install lynx -y
```

3 Clean and simplify the text

You can remove extra spaces, control characters or junk formatting:

```
cat email_body.txt | tr -d '\r' | sed '/^$/d' > email_body_clean.txt
```

- `tr -d '\r'` → removes carriage returns
- `sed '/^$/d'` → removes blank lines

- Output: **email_body_clean.txt** — clean and readable text body of the email
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Step 1: Extract Unique Words / Suspicious Tokens

Command:

```
aspell list < email_body_clean.txt | sort -u
```

Purpose: Identify unusual or suspicious words in the email that do not appear in a standard dictionary.

Output (excerpt):

```
Antispam
AuthAs
AuthSource
AxRVpRVERCZVlaQ
BCL
BFZWkvM
BOEZQSjdNSFF
...
phishing
no-reply
bounce
```

Analysis: Several tokens indicate unusual or auto-generated headers; words like phishing, no-reply, bounce are suspicious.

Step 2: Inspect Email Headers

Command:

```
sed -n '1,200p' email_body_clean.txt
```

Purpose: Extract the top 200 lines to analyze email headers for signs of spoofing or unusual routing.

Output (excerpt):

```
Content-Transfer-Encoding: 8bit
Message-ID: <996dcbd8-3d55-4865-93d2-d8471d7db546@DM6NAM12FT029.eop-
nam12.prod.protection.outlook.com>
Return-Path: bounce@nonkfrgr.co.uk
X-MS-Exchange-Organization-AuthAs: Anonymous
```

X-MS-Exchange-Organization-AuthSource: DM6NAM12FT029.eop-nam12.prod.protection.outlook.com

...
Microsoft account
Unusual sign.in activity
We detected something unusual about a recent sign-in to the Microsoft account phishing@pot.

...
Country/region: Russia/Moscow
IP address: 103.225.77.255

Analysis: Headers show multiple unusual indicators — Return-Path domain mismatch, anonymous authentication and message routing from non-official sources.

Step 3: Extract Human-Readable Content

Command:

```
cat email_body_clean.txt | sed 's/=r\n//g' > human_text.txt
```

Purpose: Remove email encoding artifacts to produce readable text.

Output (excerpt):

Microsoft account
Unusual sign.in activity
We detected something unusual about a recent sign-in to the Microsoft account phishing@pot.
Sign-in details
Country/region: Russia/Moscow
IP address: 103.225.77.255
A user from Russia/Moscow just logged into your account from a new device
...

Step 4: Filter Key Phishing Keywords

Command:

```
grep -iE 'sign[- ]?in|account|report|opt out|click here|unusual|Russia|IP|device' human_text.txt
```

Purpose: Highlight sections of the email that indicate phishing attempts.

Output:

Microsoft account
Unusual sign.in activity

We detected something unusual about a recent sign-in to the Microsoft account phishing@pot.

Sign-in details

Country/region: Russia/Moscow

IP address: 103.225.77.255

A user from Russia/Moscow just logged into your account from a new device, If this wasn't you, please report the user.

To opt out or change where you receive security notifications, click
The Microsoft account team

Analysis: Contains suspicious references to account access, unusual locations and calls to action — common phishing traits.

Step 5: Check Sender Spoofing

Command:

```
grep -i "from:" email_body_clean.txt
```

Output:

From: no-reply@access-accsecurity.com

Analysis: Sender domain is **not Microsoft** — a classic spoofing tactic.

Step 6: Check Reply-To / Return-Path

Command:

```
grep -i "return-path" email_body_clean.txt
```

Output:

Return-Path: bounce@nonkfrgr.co.uk

Analysis: Return-Path does not match Microsoft — confirms spoofed source.

Step 7: Check for Malicious Links or Urgent Wording

Command:

```
grep -iE 'click|report|opt out|verify' human_text.txt
```

Output:

Report The User

To opt out or change where you receive security notifications, click

Analysis: Urgent calls to action and suspicious links are indicators of phishing.

Step 8: Summarize Phishing Indicators

Checkpoint	Finding	Status	Severity
1. Sender spoofing	no-reply@access-accsecurity.com pretending to be Microsoft	🚩 Fake domain	High
2. Return-Path	bounce@nonkfrgr.co.uk — unrelated domain	🚩 Spoofed	High
3. Unusual activity	“Unusual sign-in activity” from a new device	⚠️ Warning	Medium
4. Country / IP	Russia/Moscow, IP 103.225.77.255 — unexpected location	🚩 Suspicious	High
5. Urgent call-to-action	“Report the user” / “Click here” to opt out or change settings	🚩 Likely phishing	High
6. Account mentioned	phishing@pot — non-official Microsoft account	🚩 Red flag	High
7. Grammar / formatting	Minor formatting issues, e.g., “sign.in” instead of “sign-in”	⚠️ Common in phishing	Low
8. Final assessment	Multiple indicators of spoofing, urgency and suspicious links	❌ Malicious	Critical

Overall Severity: Critical

Recommendation: Do not click any links, block the sender/domain and report phishing to Microsoft.

✅ MITIGATION

1. Block the sender in your email client.
2. Report phishing to Microsoft via <https://reportphishing.microsoft.com>
3. Verify account activity directly on Microsoft’s official site.
4. Avoid downloading attachments or clicking links from suspicious emails.